DATA SCIENCE PROJECT REPORT

Credit Card Fraud Detection using Machine Learning

Fraud Analysis • Machine Learning Classification • Banking Analytics

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Executive Summary

This project develops a machine learning model to detect fraudulent credit card transactions. The notebook includes data loading, exploratory analysis, preprocessing, Random Forest model training, and evaluation using accuracy, precision, recall, F1-score, MCC, and confusion matrix.

1. Introduction

Credit card fraud causes significant financial losses. The objective is to build a classifier capable of distinguishing legitimate and fraudulent transactions from historical transaction data.

2. Project Highlights

Dataset: Credit Card Fraud Detection Dataset
Rows: 284,807
Features: 30
Target: Class (0=Normal, 1=Fraud)
Algorithm: Random Forest Classifier.

3. Dataset Overview

The dataset is highly imbalanced, with fraudulent transactions representing a very small proportion of the data. Exploratory analysis and visualization are used to understand the distribution and relationships among variables.

4. Data Preprocessing

Feature selection, train-test split, handling of missing values (where applicable), and preparation of features for model training.

5. Machine Learning Model

A Random Forest Classifier is trained to classify transactions as genuine or fraudulent. Random Forest is robust, handles nonlinear relationships, and performs well on tabular data.

6. Model Evaluation

The notebook evaluates the trained model using Accuracy, Precision, Recall, F1-score, Matthews Correlation Coefficient (MCC), and a Confusion Matrix.

7. Key Findings

The project demonstrates that machine learning can effectively identify fraudulent transactions while maintaining strong performance on legitimate transactions. Special attention is required because the dataset is highly imbalanced.

8. Technical Skills Demonstrated

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Random Forest, Data Preprocessing, Model Evaluation.

9. Future Scope

Experiment with XGBoost, LightGBM, SMOTE for imbalance handling, hyperparameter tuning, SHAP explainability, and real-time deployment.

10. Project Information

Project: Credit Card Fraud Detection using Machine Learning
Repository: princemaheta2627-glitch/CraditCard_Fraud_Detection_ML
Notebook: cradit_card_fraud_detection_ML_Answer.ipynb